

# Corrections to “Notes on Thermophoretic Force”

from v21 to v22

September 2026

## Summary

Every statement of C. Chin’s *Notes on Thermophoretic Force* (v21) that can be checked was tested in the Mathematica notebook `ThermophoreticForce_v21_verification.nb`. The notebook reproduces the notes section by section and runs 106 checks. The checks use:

- symbolic algebra and linear stability analysis;
- group-theoretic symmetry reduction;
- unit analysis;
- direct numerical integration of the equations of motion;
- an independent kinetic-theory derivation of the free-molecular surface stresses, which reproduces Waldmann’s thermophoretic force and Epstein’s drag for a sphere.

Of the 106 checks, 65 confirm the notes as written, 28 found discrepancies, and 13 are remarks.

The discrepancies are collected below as 19 changes, labelled C1–C19. The same labels appear in the margin of the corrected notes, `ThermophoreticForce_v22.pdf`, next to each changed passage. Equation numbers are unchanged from v21. Equations added in v22 carry tags such as (29a) and (41a).

The corrections that change results are the following:

- **C11:** the stability analysis of the spin eigenmotions.
- **C13:** the drag anisotropy was kept by mistake in the levitation gradient and the orbit radius.
- **C14–C17:** sign and prefactor errors in the surface-element, disk and cylinder formulas.
- **C19:** the free-molecule rotational drag coefficient is 2.8 times larger than stated, which lowers the gyrophoretic estimate from  $1.6 F_T$  to  $0.56 F_T$ .

The overall picture of the notes is unchanged.  $F_T$  levitates the particle; a generic particle spins about the gradient axis and moves on a horizontal circle; and the gyrophoretic force of a strongly non-spherical particle is comparable to  $F_T$  in the transition regime.

| ID  | Location                | Type            | Change                                                                                                                                                                   |
|-----|-------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1  | Sec. 1 intro            | reference       | Disk and cylinder are in Sections 4 and 5, not 2 and 3.                                                                                                                  |
| C2  | Sec. 1.2                | clarification   | Symmetry of $\mathbf{f}_{\text{th}}$ is an assumption; symmetry of the drag tensors follows from reciprocity, not from dissipation.                                      |
| C3  | Eq. (10)                | error           | The block response multiplies $\mathbf{T}_{\text{drag}}$ by $\mathbf{v}$ ; it needs three driving fields $(\nabla \ln T, \mathbf{v}, \boldsymbol{\omega})$ .             |
| C4  | Secs. 1.4, 3            | error (wording) | $\nabla T$ along $-\hat{Z}$ means hot below, cold above.                                                                                                                 |
| C5  | Sec. 1.7                | reference       | Circular orbits are derived in Sections 2.2–2.3.                                                                                                                         |
| C6  | Sec. 1.7                | typo            | Three mirror planes give 9 DOF, not 18.                                                                                                                                  |
| C7  | Eq. (29), Table 1       | error           | $\mathbf{T}_{\text{th}} = 0$ holds for $D_{\infty h}$ (cylinders, spheroids), not for $C_{\infty v}$ , which allows an aligning torque. Added Eq. (29a) and a table row. |
| C8  | Eqs. (30)–(31)          | error (units)   | $\gamma_0 = f_0 = \eta \kappa \ell^2 T / \bar{v}$ ; the factor $T$ was missing.                                                                                          |
| C9  | Sec. 2 intro            | typo            | Four tensors with 27 parameters, not five with 24.                                                                                                                       |
| C10 | Sec. 2.1, Sec. 6        | clarification   | Gyrophoretic note completed; the effect is $O(\varepsilon^2)$ for near-spheres.                                                                                          |
| C11 | Eq. (41), Sec. 2.2      | error           | Rate formula replaced by the exact linear stability analysis, (41)–(41b); contradictory text removed; attractor rule for general spectra.                                |
| C12 | Eq. (42)                | clarification   | Neglected centripetal acceleration and the exact correction factor.                                                                                                      |
| C13 | Eqs. (44)–(49)          | error           | Drag anisotropy enters only at $O(\varepsilon^2)$ : $\delta \mathbf{A} = \delta \mathbf{f}_{\text{th}} / \xi_0$ ; fixes in (46), (47), (49) and the hover condition.     |
| C14 | Eq. (50), Sec. 3        | error           | Signs of the tangential drag and tangential thermophoretic stress; definition of $\mathbf{v}$ and $\nu$ .                                                                |
| C15 | Eq. (51)                | error           | Same tangential signs; spurious factor $a$ on the normal drag removed.                                                                                                   |
| C16 | Eqs. (52)–(54)          | error           | Prefactor $f = mN\sqrt{\pi} rL / \sqrt{h}$ ; tilt sense stated; $\theta = 0$ is a horizontal cylinder.                                                                   |
| C17 | Eq. (56)                | error           | $2\sqrt{\pi} mN\nu / \sqrt{h} = \eta \kappa T / \bar{v}$ .                                                                                                               |
| C18 | Eq. (58)                | clarification   | Waldmann’s formula involves the translational conductivity.                                                                                                              |
| C19 | Eqs. (60)–(62), Table 2 | error           | Free-molecule $C_r = \frac{8\sqrt{\pi}}{3} \frac{P}{\bar{v}} \ell^4$ ; updated $\omega_{\text{ss}}$ , $F_\omega$ , (62), Table 2 and conclusions.                        |

## Details

### C1 Section 1, first paragraph

**v21:** “The disk and cylinder examples in Sections 2 and 3 ...”

**v22:** “... in Sections 4 and 5 ...”

**Reason:** Section 2 is the near-sphere theory and Section 3 the surface element.

### C2 Section 1.2: symmetry of the tensors

**v21:**  $\mathbf{f}_{\text{th}}$  “is symmetric positive-definite (6 DOF)”; for  $\mathbf{f}_{\text{drag}}$ , “ $P < 0$  for all  $\mathbf{v}$  forces positive-definiteness, and only the symmetric part contributes to  $P$ ”.

**v22:**  $\mathbf{f}_{\text{th}}$  is *taken* symmetric, and this is stated as an assumption. The drag tensors are symmetric by Onsager/Lorentz reciprocity and positive-definite by dissipation.

**Reason:** The quadratic form  $\mathbf{v}^\top \mathbf{A} \mathbf{v}$  is blind to the antisymmetric part of  $\mathbf{A}$ , so dissipation constrains only the symmetric part. Symmetry itself is a reciprocity statement. For  $\mathbf{f}_{\text{th}}$  there is no such statement, and the chiral point group  $C_\infty$  allows an antisymmetric part: the invariant-tensor count gives 3 DOF for a general  $\mathbf{f}_{\text{th}}$ , against 2 for a symmetric one.

### C3 Eq. (10): collected linear response

**v21:**

$$\begin{pmatrix} \mathbf{F}^b \\ \boldsymbol{\tau}^b \end{pmatrix} = - \begin{pmatrix} \mathbf{f}_{\text{th}} & \mathbf{f}_{\text{drag}} \\ \mathbf{T}_{\text{th}} & \mathbf{T}_{\text{drag}} \end{pmatrix} \begin{pmatrix} (\nabla \ln T)^b \\ \mathbf{v}^b \end{pmatrix}, \quad \text{“the driving fields are } \nabla \ln T \text{ and } \mathbf{v} \text{ (not } \boldsymbol{\omega})\text{”}.$$

**v22:**

$$\begin{pmatrix} \mathbf{F}^b \\ \boldsymbol{\tau}^b \end{pmatrix} = - \begin{pmatrix} \mathbf{f}_{\text{th}} & \mathbf{f}_{\text{drag}} & 0 \\ \mathbf{T}_{\text{th}} & 0 & \mathbf{T}_{\text{drag}} \end{pmatrix} \begin{pmatrix} (\nabla \ln T)^b \\ \mathbf{v}^b \\ \boldsymbol{\omega}^b \end{pmatrix}.$$

The zero blocks are identified as the omitted translation–rotation couplings.

**Reason:** Multiplying out the v21 matrix gives  $\boldsymbol{\tau}^b = -\mathbf{T}_{\text{th}}(\nabla \ln T)^b - \mathbf{T}_{\text{drag}}\mathbf{v}^b$ . That disagrees with Eqs. (7) and (9), and it is not even dimensionally a torque.

### C4 Direction of the gradient (Sections 1.4 and 3)

**v21:** Sec. 1.4: “ $\nabla T$  is also taken along  $-\hat{Z}$  (hot above cold)”. Sec. 3: “ $\nabla T = -T'(0, 0, 1)$ , so the gas is hotter at large  $z$  (cold below, hot above)”.

**v22:** “hot below, cold above” and “colder at large  $z$ ”.

**Reason:**  $T(z) = T_0 - T'z$  decreases with height. The stated direction of the force (up, toward cold) was already correct.

### C5 Section 1.7, Level 1

**v21:** “...circular orbits (Section 2.2).”

**v22:** “(Sections 2.2–2.3).”

**Reason:** The orbit itself is derived in Section 2.3.

### C6 Section 1.7, Level 2 heading

**v21:** “Three mutually perpendicular mirror planes (18 DOF).”

**v22:** “(9 DOF).”

**Reason:** Invariance under the three reflections leaves  $3 + 3 + 0 + 3 = 9$  parameters, as the text below Eq. (28) and Table 1 already state.

### C7 Section 1.7, Level 3a and Table 1

**v21:** “Uniaxial symmetry,  $C_{\infty v}$  (6 DOF) ...  $\mathbf{T}_{\text{th}} = 0$ . Bodies: cylinders, spheroids.”

**v22:** “Uniaxial with a perpendicular mirror plane,  $D_{\infty h}$  (6 DOF)”, followed by a new paragraph and Eq. (29a):

$$C_{\infty v} : \quad \mathbf{T}_{\text{th}} = \lambda_a [\hat{e}_3]_\times, \quad \boldsymbol{\tau}_{\text{th}}^b = -\lambda_a \hat{e}_3 \times (\nabla \ln T)^b \quad (7 \text{ DOF}).$$

Table 1 gains a row for  $C_{\infty v}$  and labels the uniaxial row  $D_{\infty h}$ .

**Reason:**  $\mathbf{T}_{\text{th}}$  is a pseudotensor,  $X = \det(g) g X g^\top$ . Solving the invariance conditions gives one invariant for  $C_{\infty v}$ : a cone or Janus particle feels a torque that aligns its axis with the gradient.  $\mathbf{T}_{\text{th}}$  vanishes only for  $D_{\infty h}$ , which contains the perpendicular mirror plane; cylinders and spheroids belong to this group. The form of the other three tensors is the same for both groups.

## C8 Eqs. (30)–(31): sphere value of $\mathbf{f}_{\text{th}}$

**v21:**  $\gamma_0 = \eta(\kappa/\bar{v})\ell^2$ .

**v22:**  $\gamma_0 = \eta(\kappa/\bar{v})\ell^2 T$  (and  $f_0 = \eta\kappa\ell^2 T/\bar{v}$  in the Waldmann remark after Eq. (30)).

**Reason:**  $\mathbf{f}_{\text{th}}$  multiplies  $\nabla \ln T = \nabla T/T$  and has units of N m (Eq. (11)).  $\eta\kappa\ell^2/\bar{v}$  has units J/K. Waldmann’s force, Eq. (58), is written with  $\nabla T$ , and converting it supplies the factor  $T$ .

## C9 Section 2, introduction

**v21:** “five tensors with up to 24 independent parameters”.

**v22:** “four tensors with up to 27 independent parameters”.

**Reason:** See Table 1.

## C10 Gyrophoretic coupling (Sections 2.1 and 6)

**v21:** “Since no experimental evidence currently constrains the relationship and  $\mathbf{T}_{\text{th}}$ ; any gyrophoretic coupling is  $O(\varepsilon^2)$  and is neglected.”

**v22:** The sentence is completed. For a near-sphere the coupling is  $O(\varepsilon)$  and the steady spin is  $O(\varepsilon)$ , so the gyrophoretic force is  $O(\varepsilon^2)$ . Section 6 now states that its estimate refers to a strongly non-spherical particle ( $\varepsilon \sim 1$ ) and carries a factor  $\varepsilon^2$  for near-spheres.

**Reason:** The v21 sentence was missing a word. Section 6 concluded that the effect is comparable to  $F_T$  without reconciling this with its neglect in Section 2.

## C11 Eq. (41): stability of the three eigenmotions

**v21:**

$$\sigma_{kj} = \frac{G\varepsilon(\lambda_k - \lambda_j)\lambda_j}{I_0 c_0} \quad (j \neq k).$$

The text first concludes that for  $\lambda_3 > \lambda_2 > \lambda_1 > 0$  “the stable attractor is then  $\mathbf{e}_1$ ”, and then that  $\mathbf{e}_3$  is the stable attractor (table). It states that “all generic initial conditions converge to eigenmotion 3”.

**v22:** Exact linearisation of Eq. (37) about  $(\boldsymbol{\omega}^b, \mathbf{u}) = (g\lambda_k \mathbf{e}_k/c_0, \mathbf{e}_k)$ , with  $g = G\varepsilon$ . The transverse modes obey the quartic

$$c_0^2 I_0^2 s^4 + 2c_0^3 I_0 s^3 + (c_0^4 + g^2 I_0^2 \lambda_k^2) s^2 + c_0 g^2 I_0 \lambda_k (2\lambda_k - \lambda_j - \lambda_l) s + c_0^2 g^2 (\lambda_k - \lambda_j)(\lambda_k - \lambda_l) = 0,$$

whose Routh–Hurwitz conditions are exact. For weak inertia,

$$s \simeq \pm i \frac{g}{c_0} \sqrt{(\lambda_k - \lambda_j)(\lambda_k - \lambda_l)} - \frac{g^2 I_0}{2c_0^3} [\lambda_j(\lambda_k - \lambda_l) + \lambda_l(\lambda_k - \lambda_j)],$$

so  $\mathbf{e}_k$  is stable if and only if (a)  $(\lambda_k - \lambda_j)(\lambda_k - \lambda_l) > 0$  and (b)  $\lambda_j(\lambda_k - \lambda_l) + \lambda_l(\lambda_k - \lambda_j) > 0$ . The table of v21 ( $\mathbf{e}_3$  stable) is kept, restricted to a positive-definite  $\delta\mathbf{T}_{\text{th}}$ . For a negative-definite spectrum the attractor is  $\mathbf{e}_1$ ; for some mixed spectra, e.g.  $\{-1, 0.5, 2\}$ , no eigenmotion is stable.

The contradictory sentences and the two opposite definitions of  $V(\mathbf{u})$  are removed. The statement that  $V = \mathbf{u}^\top \delta \mathbf{T}_{\text{th}} \mathbf{u}$  increases monotonically is kept, for the positive-definite case, where it was confirmed numerically.

**Reason:** (41) is not dimensionally a rate: with  $[G] = 1/\text{m}$ ,  $[\varepsilon \delta \mathbf{T}_{\text{th}}] = \text{N m}^2$ ,  $[I_0] = \text{kg m}^2$ ,  $[c_0] = \text{N m s}$  it has units  $1/(\text{kg m}^2 \text{ s})$ . Alignment is a second-order effect that requires the lag  $I_0/c_0$ . Its rate is  $\sim g^2 I_0 \lambda^2 / c_0^3$ , while the precession frequency is  $\sim g \lambda / c_0$ . Nonlinear integration of (37) from random initial conditions confirms each case (positive spectrum  $\rightarrow \mathbf{e}_3$ , negative  $\rightarrow \mathbf{e}_1$ , mixed  $\rightarrow$  no fixed point).

## C12 Eq. (42): neglected centripetal acceleration

**v22:** Added: for a precessing body  $\dot{\mathbf{v}} = \boldsymbol{\Omega} \times \mathbf{v} \neq 0$ . Neglecting it requires  $M\Omega/\xi_0 \ll 1$ , and the exact circular solution is reduced by  $[1 + (M\Omega_3/\xi_0)^2]^{-1/2}$ .

**Reason:** With this factor the quasi-steady radius agrees with direct integration of Eqs. (23)–(26). The residual difference shrinks like  $\varepsilon^2$ .

## C13 Eqs. (44)–(49): levitation gradient and orbit radius

**v21:**

$$v_Z = \frac{\gamma_0 G - Mg}{\xi_0} + \varepsilon G (\delta \mathbf{A})_{33}, \quad \delta \mathbf{A} \equiv \frac{1}{\xi_0} (\delta \mathbf{f}_{\text{th}} - A_0 \delta \mathbf{f}_{\text{drag}}), \quad G = \frac{Mg}{\gamma_0 + \varepsilon \xi_0 (\delta \mathbf{A})_{33}} = \frac{Mg}{\gamma_0} [1 - \varepsilon A_0 (\delta \mathbf{A})_{33}],$$

$$\mathbf{v}^b = \varepsilon G_0 \delta \mathbf{A} \mathbf{u} - \varepsilon G_0 A_0 (\delta \mathbf{A})_{33} \mathbf{u}.$$

**v22:**

$$\delta \mathbf{A} \equiv \frac{1}{\xi_0} \delta \mathbf{f}_{\text{th}}, \quad G = \frac{Mg}{\gamma_0 + \varepsilon \xi_0 (\delta \mathbf{A})_{33}} = \frac{Mg}{\gamma_0} \left[ 1 - \varepsilon \frac{(\delta \mathbf{A})_{33}}{A_0} \right] = \frac{Mg}{\gamma_0 + \varepsilon (\delta \mathbf{f}_{\text{th}})_{33}},$$

$$\mathbf{v}^b = \varepsilon G_0 \delta \mathbf{A} \mathbf{u} - \varepsilon G_0 (\delta \mathbf{A})_{33} \mathbf{u} = \varepsilon G_0 \Pi_u \delta \mathbf{A} \mathbf{u}, \quad R_{\text{orbit}} = \frac{c_0 |\Pi_{\mathbf{e}_3} \delta \mathbf{f}_{\text{th}} \mathbf{e}_3|}{\xi_0 |\lambda_3|}.$$

The hover condition (key feature 3) becomes “ $\mathbf{e}_3$  is also an eigenvector of  $\delta \mathbf{f}_{\text{th}}$ ”.  $\delta \mathbf{f}_{\text{drag}}$  plays no role. In Case 1 the drift is  $\varepsilon G_0 R \Pi_u \delta \mathbf{A} R^\top \hat{Z}$ .

**Reason:** Eq. (43) (correct) multiplies  $\delta \mathbf{f}_{\text{drag}}$  by  $(\gamma_0 G - Mg)$ , which is  $O(\varepsilon)$  near levitation. The drag term is therefore  $O(\varepsilon^2)$ . Definition (45) had replaced  $(\gamma_0 G - Mg)$  by  $\gamma_0 G$ . Expanding the exact levitation condition  $\mathbf{u} \cdot \mathbf{f}_{\text{drag}}^{-1} (G \mathbf{f}_{\text{th}} - Mg) \mathbf{u} = 0$  in  $\varepsilon$  gives  $G = Mg/(\gamma_0 + \varepsilon \delta f_{\text{th},33})$ . In v21, the two forms in (46) disagree unless  $A_0 = 1$ , and (47) gives  $\mathbf{u} \cdot \mathbf{v}^b = \varepsilon G_0 (\delta \mathbf{A})_{33} (1 - A_0) \neq 0$ , contrary to the text below it.

*Numerical confirmation.* For a random near-sphere with  $\varepsilon = 0.01$ , direct integration of Eqs. (23)–(26) on the steady precession gives  $R_{\text{orbit}} = 0.2413$ . The corrected (49) gives 0.2403, a 0.4% difference that shrinks with  $\varepsilon$ . The v21 formula gives 0.1271 (47% off), and its error does not shrink with  $\varepsilon$ .

## C14 Eq. (50): stress on a surface element

**v21:**

$$\frac{1}{A} \begin{pmatrix} F_{t_1} \\ F_{t_2} \\ F_n \end{pmatrix} = \alpha \begin{pmatrix} 0 \\ 0 \\ -\frac{1}{2} \sqrt{\pi/h} \end{pmatrix} - \alpha \begin{pmatrix} -\frac{a}{2} \mathbf{v} \cdot \hat{t}_1 \\ -\frac{a}{2} \mathbf{v} \cdot \hat{t}_2 \\ [2 - a(1 - \pi/4)] \mathbf{v} \cdot \hat{n} \end{pmatrix} + \alpha \frac{3\nu}{4} \frac{T'}{T} \begin{pmatrix} \frac{a}{2} \sin \theta \\ 0 \\ (2 - a) \cos \theta \end{pmatrix}.$$

**v22:**

$$\frac{1}{A} \begin{pmatrix} F_{t_1} \\ F_{t_2} \\ F_n \end{pmatrix} = \alpha \begin{pmatrix} 0 \\ 0 \\ -\frac{1}{2}\sqrt{\pi/h} \end{pmatrix} - \alpha \begin{pmatrix} \frac{a}{2}\mathbf{v} \cdot \hat{t}_1 \\ \frac{a}{2}\mathbf{v} \cdot \hat{t}_2 \\ [2 - a(1 - \pi/4)]\mathbf{v} \cdot \hat{n} \end{pmatrix} + \alpha \frac{3\nu}{4} \frac{T'}{T} \begin{pmatrix} -\frac{a}{2} \sin \theta \\ 0 \\ (2 - a) \cos \theta \end{pmatrix},$$

with  $\mathbf{v}$  the element velocity relative to the gas and  $\nu = \frac{4}{15}\kappa/(Nk_B)$ , which is the kinematic viscosity for a monatomic gas.

**Reason:** These stresses were derived independently from a Grad (13-moment) distribution carrying the heat flux  $\mathbf{q} = \kappa T' \hat{z}$ , with fraction  $1 - a$  of the re-emission specular and fraction  $a$  diffuse at the gas temperature. The derivation reproduces the pressure, the normal drag, and the  $2(2 - a)/a$  ratio of the thermophoretic stresses exactly. Both tangential terms have the opposite sign to v21: the tangential drag opposes the motion, and the tangential thermophoretic stress points along  $\mathbf{q}$ , toward the cold side. Integrated over a sphere, the corrected stresses give Epstein's drag  $\frac{4\pi}{3}R^2\rho\bar{c}(1 + \pi a/8)$  and Waldmann's force  $\frac{16\sqrt{\pi}}{15}R^2\kappa T'/\bar{v}$ , independent of  $a$ . With the v21 signs the sphere force would be  $(1 - a)$  times Waldmann's and would vanish for diffuse reflection. The v21 cylinder formulas (52)–(53) are consistent only with the corrected signs.

#### C15 Eq. (51): disk

**v21:**  $\alpha a(\mathbf{v} \cdot \hat{t}_1, \mathbf{v} \cdot \hat{t}_2, -2[2 - a(1 - \pi/4)]\mathbf{v} \cdot \hat{n}) + \alpha \frac{3\nu}{4} \frac{T'}{T} (a \sin \theta, 0, 2(2 - a) \cos \theta)$ .

**v22:**  $-\alpha(a\mathbf{v} \cdot \hat{t}_1, a\mathbf{v} \cdot \hat{t}_2, 2[2 - a(1 - \pi/4)]\mathbf{v} \cdot \hat{n}) + \alpha \frac{3\nu}{4} \frac{T'}{T} (-a \sin \theta, 0, 2(2 - a) \cos \theta)$ .

**Reason:** This is the sum of the corrected Eq. (50) over the faces  $\pm\hat{n}$ . The tangential signs follow C14. The normal drag doubles to  $-2\alpha[2 - a(1 - \pi/4)]\mathbf{v} \cdot \hat{n}$ , without the extra factor  $a$  of v21.

#### C16 Eqs. (52)–(54): cylinder

**v21:**  $f = mN\pi rL/\sqrt{h}$ ; axis “tilted at angle  $\theta$  from the  $X$ -axis”; “For  $v = 0$  and  $\theta = 0$  (cylinder axis along  $Z$ ) the levitation force is ...”.

**v22:**  $f = mN\sqrt{\pi}rL/\sqrt{h} = \alpha\pi rL$ . The axis is  $\hat{a} = (\cos \theta, 0, -\sin \theta)$ , tilted below the  $X$ -axis, and the opposite tilt flips both cross terms.  $\theta = 0$  is a horizontal cylinder, for which  $F_Z = fBT'/T$ ; a vertical cylinder feels  $F_Z = fDT'/T$ . End caps are neglected ( $L \gg r$ ).

**Reason:** Integrating the corrected Eq. (50) over the lateral surface reproduces all four coefficients  $A, B, C, D$  of (52)–(53) exactly, with the prefactor  $\alpha\pi rL$ . That is  $1/\sqrt{\pi}$  times the v21  $f$ . Eq. (55) is confirmed. Cross-check: the same stresses give the Garcia-Ybarra & Rosner sphero-cylinder results quoted in Zheng (2002): about +31% drift speed parallel to the gradient at  $L/r \approx 15$ , a few per cent slower perpendicular, and up to  $12^\circ$  deflection.

#### C17 Eq. (56): conversion between conventions

**v21:**  $\frac{2mN\pi\nu}{\sqrt{h}} = \frac{\eta\kappa}{\bar{v}}$ .

**v22:**  $\frac{2\sqrt{\pi}mN\nu}{\sqrt{h}} = \frac{\eta\kappa T}{\bar{v}}$ .

**Reason:** The v21 left side has units  $\text{kg/s}^2$  and the right side  $\text{kg}/(\text{s}^2 \text{K})$ . The corrected relation follows from equating the sphere force in both conventions,  $F_o = 2\pi\alpha\nu R^2 T'/T = \eta\kappa R^2 T'/\bar{v}$ , and it holds identically with the kinetic value of  $\nu$ .  $\eta = 16\sqrt{\pi}/15$  is confirmed.

## C18 Eq. (58): conductivity in Waldmann's formula

**v22:** Added a remark: Waldmann's result involves the translational conductivity,  $\kappa_{\text{tr}} = \frac{15}{4}(k_B/m)\mu \approx 0.020 \text{ W/(m K)}$  for air. Using it instead of the total  $\kappa = 0.026 \text{ W/(m K)}$  would lower  $F_T$  by about 25%.

## C19 Eqs. (60)–(62), Table 2 and conclusions: rotational drag and gyrophoretic force

**v21:**  $C_r = (8\pi/15)(P/\bar{v})\ell^4 \approx 4.3 \times 10^{-19} \text{ N m s}$ ;  $\omega_{\text{ss}} \approx 4.4 \times 10^4 \text{ rad/s}$ ;  $F_\omega \approx 1.5 \times 10^{-9} \text{ N}$ ;

$$\frac{F_\omega}{F_T} = \frac{\eta\kappa T}{\bar{v}} \frac{8\pi}{15P\ell} \approx 3Kn \approx 1.6.$$

**v22:**  $C_r = \frac{2\pi}{3}\rho\bar{c}\ell^4 = \frac{8\sqrt{\pi}}{3}\frac{P}{\bar{v}}\ell^4 \approx 1.2 \times 10^{-18} \text{ N m s}$ ;  $\omega_{\text{ss}} \approx 1.6 \times 10^4 \text{ rad/s}$  ( $\approx 7.5 \times 10^{-4}\bar{v}/\ell$ );  $F_\omega \approx 5.3 \times 10^{-10} \text{ N}$ ;

$$\frac{F_\omega}{F_T} = \frac{\eta\kappa T}{\bar{v}} \frac{3}{8\sqrt{\pi}P\ell} \approx 0.56 = \frac{45\eta}{32\pi}Kn \quad (\kappa = \kappa_{\text{tr}}) \approx 0.85Kn.$$

Table 2 and conclusions 2–3 are updated:  $F_\omega/F_T \approx 0.56$ ,  $\omega_{\text{ss}}\ell/\bar{v} \approx 8 \times 10^{-4}$ , and  $v/\bar{v} \approx 3 \times 10^{-4}$  with Epstein drag.

**Reason:** For diffuse re-emission, each molecule striking a rotating sphere leaves with the local wall velocity. The resulting tangential momentum flux  $\frac{1}{4}n\bar{c}m\mathbf{V}_{\text{wall}}$  gives the torque  $-\frac{2\pi}{3}\rho\bar{c}R^4\boldsymbol{\omega}$ . The same result follows from integrating the tangential drag stress of Eq. (50). This is 2.8 times the v21 coefficient. Separately, the printed analytic form of (62) has  $8\pi/15$  inverted: substituting the v21  $C_r$  into  $F_T\ell/(|\nabla \ln T|\bar{v}C_r)$  gives  $15/(8\pi P\ell)$ , which reproduces the stated 1.6. The printed  $8\pi/(15P\ell)$  would give 4.5. The qualitative conclusion stands: the gyrophoretic force of a strongly non-spherical particle is comparable to  $F_T$  in the transition regime.

## Statements confirmed without change

For completeness, these were verified as written:

- the frame conventions and orthogonality of  $R(\mathbf{q})$ , Eq. (3);
- the dimensions in (11) and the DOF count of 27;
- the equations of motion (12)–(15), including  $R\widehat{\boldsymbol{\omega}}R^\top = \widehat{R\boldsymbol{\omega}}$ ;
- the gyroscopic term (16);
- the quaternion relations (18)–(21), including that (21) reproduces (14);
- a full numerical integration of (23)–(26), with exact energy balance;
- the DOF counts of (27), (28) and (30), and the vanishing of  $\mathbf{T}_{\text{th}}$  for three mirror planes;
- Eqs. (35)–(40): the reduced dynamics and the steady eigenmotions;
- Eqs. (42)–(43);
- the horizontal circular orbit of Case 3;

- the cylinder coefficients  $A$ – $D$  and Eq. (55);
- all numerical values in (57)–(59) and (61) as computed from the v21 inputs;
- $\ell/L_T \approx 1.3 \times 10^{-3}$ .